# 3GPP Technical Specification — Sample

**Sample document based on the structure of 3GPP TS 38.331**

| Document field | Sample value |
|---|---|
| Specification | 3GPP TS 38.331 |
| Title | NR; Radio Resource Control (RRC); Protocol specification |
| Technology | 5G New Radio (NR) |
| Example release | Release 17 |
| Document status | Educational / illustrative sample |

## Disclaimer

This document is an original educational sample intended to demonstrate the organization and terminology commonly found in a 3GPP technical specification. It is not an official 3GPP document and does not reproduce normative 3GPP text.

## Contents

# 1. Scope and Terminology

## 1.1 Scope

Radio Resource Control (RRC) provides control-plane procedures between a UE and the radio access network. In NR, RRC signaling is used for tasks such as connection establishment, configuration, reconfiguration, measurement control, mobility-related procedures, and release.

This sample focuses on the conceptual structure rather than reproducing the exact normative behavior of any 3GPP release.

## 1.2 Key entities

| Term | Meaning |
|---|---|
| UE | User Equipment; the device communicating over the NR radio interface. |
| gNB | 5G NR base station providing radio access and control functions. |
| RRC | Radio Resource Control protocol layer. |
| NR | New Radio, the 5G radio access technology specified by 3GPP. |
| RAN | Radio Access Network. |

## 1.3 High-level protocol relationship

A simplified protocol stack can be represented as:

```
UE application → IP → SDAP → PDCP → RLC → MAC → PHY → NR radio interface
```

RRC operates as a control-plane protocol and interacts with lower-layer radio protocols to configure resources and behavior.

## 1.4 Normative language

Real 3GPP specifications use precise normative wording. Terms such as “shall”, “shall not”, “may”, and “should” have specific implications depending on the specification and context.

# 2. RRC Architecture and States

## 2.1 Conceptual states

For illustration, an NR UE can be viewed as transitioning between high-level RRC states according to network procedures. The exact state model and transitions are defined by the applicable 3GPP specification and release.

| State | Conceptual description |
| --- | --- |
| RRC_IDLE | UE is not maintaining an active RRC connection with a serving cell. |
| RRC_INACTIVE | UE retains certain context while reducing signaling and power consumption. |
| RRC_CONNECTED | UE has an active RRC connection and can exchange dedicated control information. |

## 2.2 Example state transition

A simplified conceptual sequence is:

```
IDLE → connection establishment → CONNECTED → suspend/release-related procedure → INACTIVE or
IDLE
```

## 2.3 Configuration concepts

RRC configuration can include radio bearers, measurement configuration, serving-cell parameters, bandwidth-part information, scheduling-related configuration, security-related configuration, and other UE-specific parameters.

## 2.4 Why RRC matters

RRC provides the control mechanisms needed to coordinate how the UE uses the radio network. Efficient RRC configuration can affect connection setup time, mobility behavior, signaling overhead, and radio resource utilization.

# 3. Example RRC Procedures

## 3.1 Simplified connection procedure

The following is an illustrative flow and is not a normative message sequence:

| Step | Direction | Illustrative action |
|---|---|---|
| 1 | UE → gNB | UE requests establishment of an RRC connection. |
| 2 | gNB → UE | Network responds with configuration for continuing the procedure. |
| 3 | UE → gNB | UE confirms completion and provides required information. |
| 4 | gNB → UE | Network applies or updates dedicated configuration. |

## 3.2 Reconfiguration concept

During an RRC reconfiguration, the network may provide updated radio parameters. The UE processes the configuration, applies applicable changes, and reports completion or failure according to the defined procedure.

## 3.3 Measurement control

The network can configure measurement-related behavior so that the UE evaluates serving and neighboring cells. Measurement results can support mobility and radio-resource decisions.

## 3.4 Failure handling

Real specifications define detailed failure cases, timers, conditions, and recovery procedures. An implementation must follow the exact applicable specification rather than relying on a simplified flow such as the one shown here.

# 4. Information Elements and Implementation Notes

## 4.1 Example information-element structure

| Information element | Example role | Example type |
|---|---|---|
| UE capability | Describes supported radio features. | Structured information |
| Cell configuration | Provides serving-cell parameters. | Configuration set |
| Measurement configuration | Defines measurement behavior. | Configuration set |
| Radio bearer configuration | Controls bearer-related parameters. | Configuration set |

## 4.2 ASN.1-style illustrative notation

```
SampleConfig ::= SEQUENCE { configId INTEGER, enabled BOOLEAN, parameterValue INTEGER }
```

The notation above is intentionally simplified and is not an extract from 3GPP TS 38.331.

# 5. Conformance and Implementation Considerations

A practical UE or gNB implementation must use the exact ASN.1 definitions, procedures, timers, conditions, state transitions, and normative requirements from the applicable 3GPP release. Implementers should also consider interoperability testing, negative cases, message validation, version compatibility, and protocol tracing.

## 5.1 Useful study approach

For learning 3GPP RRC, a useful progression is: understand the RRC states → learn major message types → study information elements → trace a connection/reconfiguration procedure → examine ASN.1 definitions → compare behavior across releases.

## End of sample

This five-page sample is provided for educational purposes.